OneEDGE – Getting Started

Application Note

1VV0301585 Rev. 12 – 2022-06-14

APPLICABILITY TABLE

PRODUCTS	Platform Version ID
ME910C1-AU ME910C1-E1 ME910C1-E2 ME910C1-NA ME910C1-NV ME910C1-WW ML865C1-EA ML865C1-NA	30
ME910G1-W1 ME910G1-WW ME310G1-W1 ME310G1-WW ME310G1-W2 ML865G1-WW	37
LE910C1-EUX LE910C1-WWX LE910C1-SAX LE910C1-SVX	25

CONTENTS

1. INTRODUCTION

1.1. Scope

This document provides a preliminary overview of the main features of OneEdge. A guideline on how to get start using the OneEdge functionalities and resources describing the methods and available functions.

1.2. Audience

The present Getting Started Guide is addressed to Users that need to learn and start using the OneEdge functionality of Telit Module.

1.3. Contact Information, Support

For general contact, technical support services, technical questions and report of documentation errors contact Telit Technical Support at:

- *TS-ONEEDGE@telit.com*

Alternatively, use:

https://www.telit.com/support

Product information and technical documents are accessible 24/7 on our web site:

https://www.telit.com

1.4. Symbol Convention

	**Danger:** This information MUST be followed or catastrophic equipment failure or personal injury may occur.
	**Warning:** Alerts the User on important steps about the module integration.
	**Note/Tip:** Provides advice and suggestions that may be useful when integrating the module.
	**Electro-static Discharge:** Notifies the User to take proper grounding precautions before handling the product.

Table 1: Symbol Conventions

2. OVERVIEW

2.1. OneEdge Description

OneEdge is a management system based on the LWM2M protocol.

OneEdge consists of an innovative module-embedded software system with pre-packaged, secure, easy-to-use deployment and management tools.

OneEdge dramatically simplifies the design, deployment and management of IoT products and solutions.

Telit's pre-provisioned OneEdge modules are ready to connect to the Telit OneEdge IoT Portal as soon as they are activated.

Figure 1: OneEdge system components

2.2. OneEdge High-Level Architecture

OneEdge can be implemented into two device architectures based on different interfaces used to manage the LwM2M client:

- Hosted Architecture via AT commands interface
- Hostless Architecture via AppZone application

Combination of both is also possible.

2.2.1. Hosted Architecture

In Hosted architecture the host MCU interfaces with the Telit module via UART or USB and controls the module and interactions with the cellular network via AT commands.

The sensors are connected directly to the MCU and the Customer application logic is located in the MCU including the communication with the external sensors (GPIO, I2C, SPI interfaces).

Specific AT commands and URCs are provided to manage LwM2M functionalities (e.g. AT#LWM2MR, AT#LWM2MW).

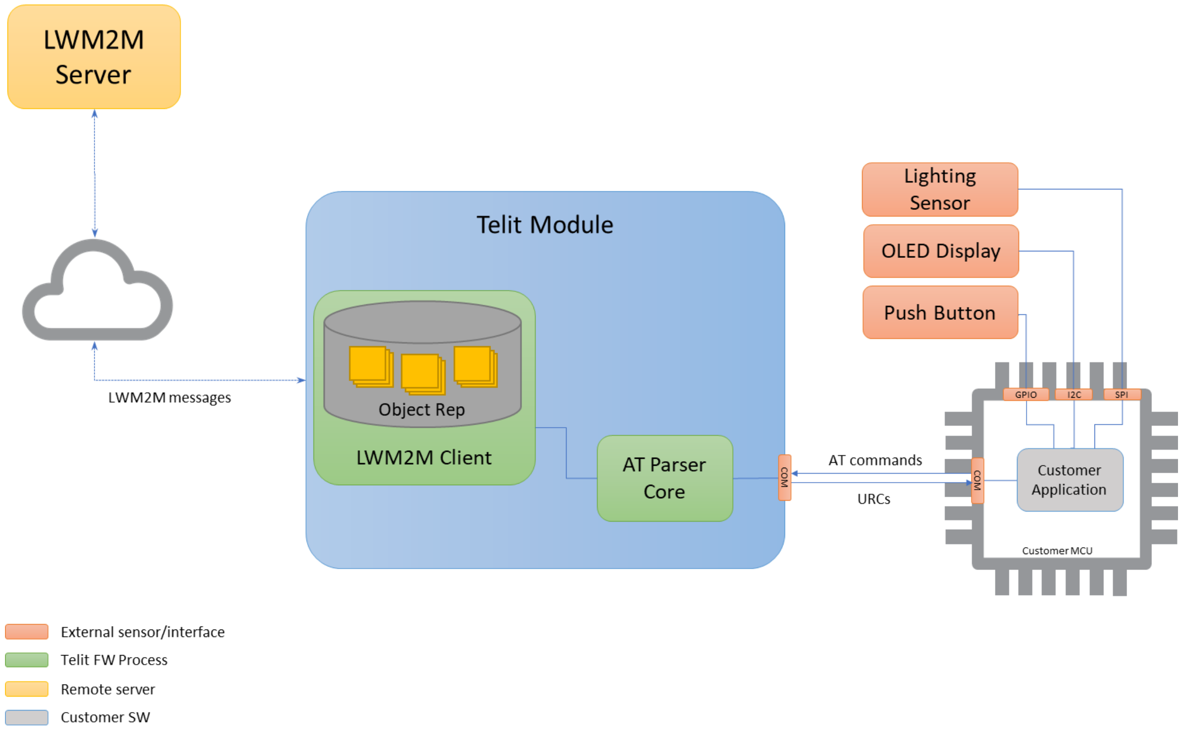


Figure 2: Hosted Architecture

2.2.2. Hostless Architecture

In case of Hostless Architecture the Telit module runs the main application in the AppZone environment. The User develops an application in C and controls the modem and interactions with the cellular network via AppZone APIs.

The Sensors are directly connected to the Telit modules through the integrated hardware interfaces (SPI, I2C etc.).

The AppZone environment provides the APIs to handle the LwM2M functionalities

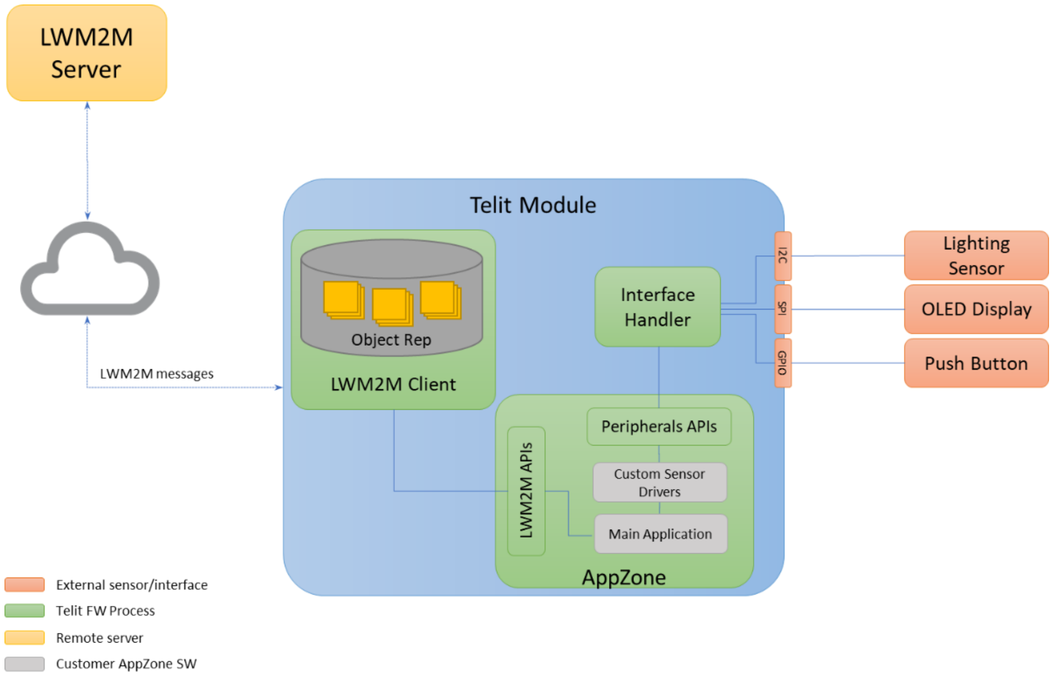


Figure 3: Hostless Architecture

2.3. LwM2M protocol

Lightweight M2M (LwM2M) is a standard protocol defined by the Open Mobile Alliance (OMA) for managing machine-to-machine (M2M) and Internet of Things (IoT) devices and enabling services.

LwM2M provides a light and compact secure communication interface along with an efficient data model.

Lightweight M2M manages the communication between an LwM2M Server, integrated on the Telit OneEdge IoT Portal, and an LwM2M client, integrated in the Telit modem. The diagram below shows the LwM2M architecture.

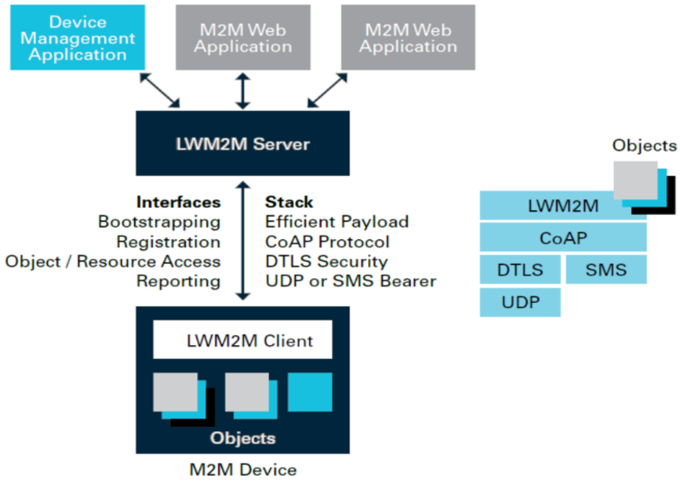


Figure 4: LwM2M Architecture

Note: The LwM2M client in Telit devices is compliant to LwM2M Version 1.0

3. ONEEDGE PREREQUISITES

3.1. ORG on OneEdge IoT Portal

Before using OneEdge, the User must have a valid organization on the OneEdge IoT Portal: on DEV portal (portal-dev.telit.com), a valid organization should be in the format OE-CUST-XXXX.

If the User doesn't have a valid org yet, he must contact the reference sales or send an email to OneEdge technical support team (ts-oneedge@telit.com).

Telit support team will create the new organization and LwM2M things on the OneEdge IoT Portal and the User will receive an invitation via email.

Warning: The User should not:

- create its own account on the OneEdge IoT Portal;
- create its own organization on the OneEdge IoT Portal;
- create LwM2M things.

3.2. Required Equipment

In order to start using OneEdge, the User must have the following equipment.

3.2.1. EVB Kit

The Telit Evaluation Board is a generic platform for developers, designed to provide the complete development environment to the User and intended to be used with a set of Telit IoT modules.

Telit EVB kit contains the following:

- Main board, it could be:
  - EVB with socket for module interface board (Figure 5)
  - EVB 2.0 with socket for module interface board

    https://www.telit.com/support-tools/development-evaluation-kits/evb2-0/
  - EVK2 with socket for module interface board (Figure 6)
  - Bravo evaluation kit

    https://www.telit.com/support-tools/development-evaluation-kits/telit-bravo-evaluation-kit/

- Charlie evaluation kit

  https://www.telit.com/support-tools/development-evaluation-kits/charlie-evaluation-kit-for-cellular-lpwa/

Figure 5: EVB (without Telit module)

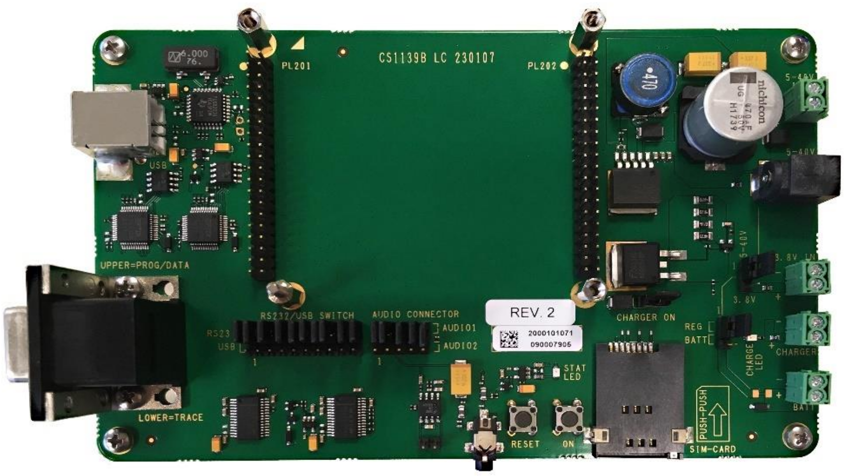


Figure 6: EVK2 (without Telit module)

- Telit Module on interface board

Figure 7: example of Interface board with Telit Module

- Power supply adapter & power cable
- Micro/Mini USB Cable or Serial Cable
- Cellular Antenna

3.2.2. SIM Card

The SIM Card provides cellular connectivity to the Telit modem.

3.2.3. PC

To connect the EVB kit to OneEdge, a PC with a browser is needed.

3.2.4. Telit AT Controller Tool

TATC Tool is the software that communicates with the Telit module and enables the User to easily execute AT commands.

The tool and all the necessary drivers can be downloaded from the following link

https://www.telit.com/evkevb-drivers/

Warning: (PC USB Drivers Installation)

Drivers are required to allow communication between the PC and the Telit module. Communication can take place either through the native USB port or the USB to serial converter (USB connector marked "FTDI"). The links provided above contains all the necessary drivers.

3.2.5. Active Account on Telit OneEdge IoT Portal

When the User logs in Telit OneEdge IoT Portal for the first time, he is asked to configure the page refreshing options to enable real-time updates on things.

Our suggestion for the first evaluation is to set it equal to 5 sec for a real-time update experience.

Warning: Since the auto refresh feature consumes many APIs on the portal billing, it's strongly recommended to disable the feature in production environment.

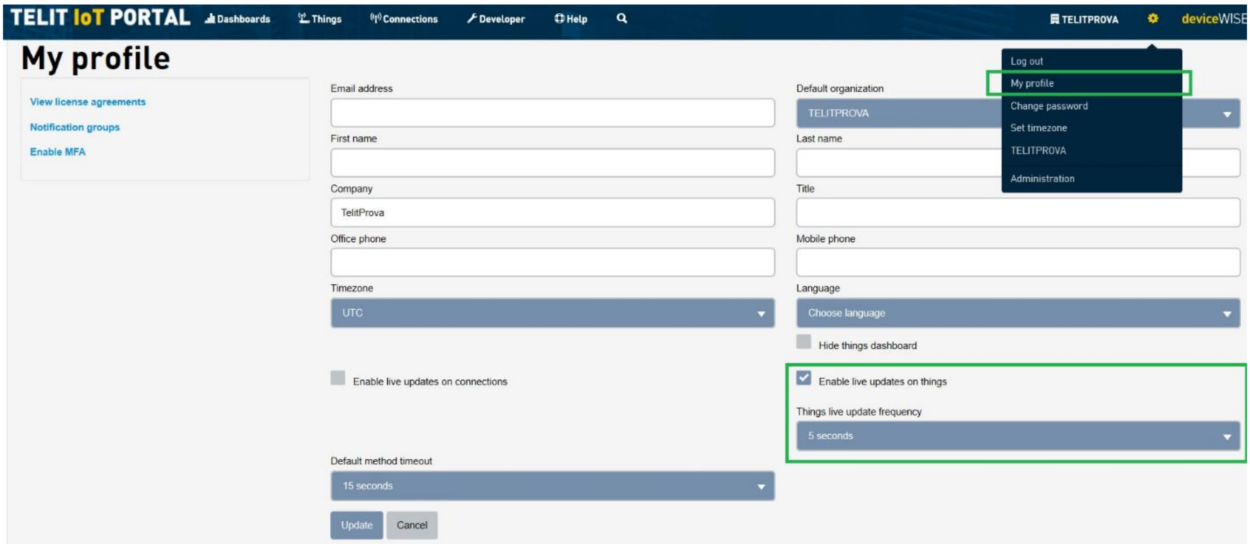


Figure 8: things live update settings

3.3. Disable WWAN Connection in Windows

Warning: (For Windows 10 and USB connection Users only). By default, the OS detects and takes control of the modem data connection. Make sure this option is disabled as described in the following paragraphs.

Whenever the modem is configured with a valid APN the Windows 10 network manager is able to detect this valid configuration and automatically start a Dial-up connection as soon as the modem is connected to the PC.

In this scenario, since the context is already enabled by Windows, the LwM2M client is unable to perform any connections with the same PDP.

To avoid this side effect, it is recommended to disable the network adapter at runtime or change the default setting in the Windows registry.

3.3.1. Disable network adapter at runtime

- Open the Device Manager Windows tool
- Open "Network adapters" section
- Search for "Telit USB WWAN Adapter" element
- Right-click and disable the device

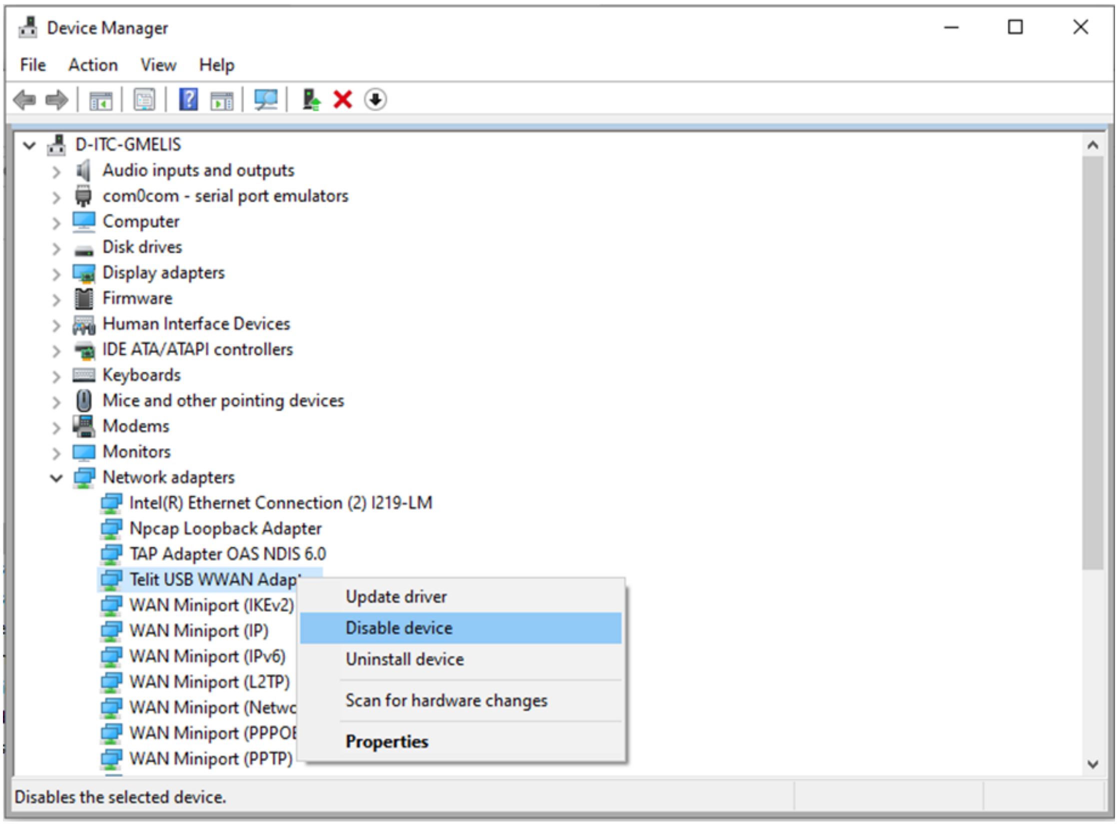


Figure 9: WWAN Connection

Warning: Note that this setting only applies to the device currently connected to the computer via USB, so the User should repeat this procedure for all new devices connected to the computer.

3.3.2. Change the Default Setting on Windows Registry

- Open the Registry Editor
- Search for:

HKEY_LOCAL_MACHINE\Software\Policies\Microsoft\Windows\WcmSvc\Local

- Right mouse click -> New -> DWORD (32-bit) value
- Name of the key must be: fMinimizeConnections
- Value set to 1
- Reboot the PC

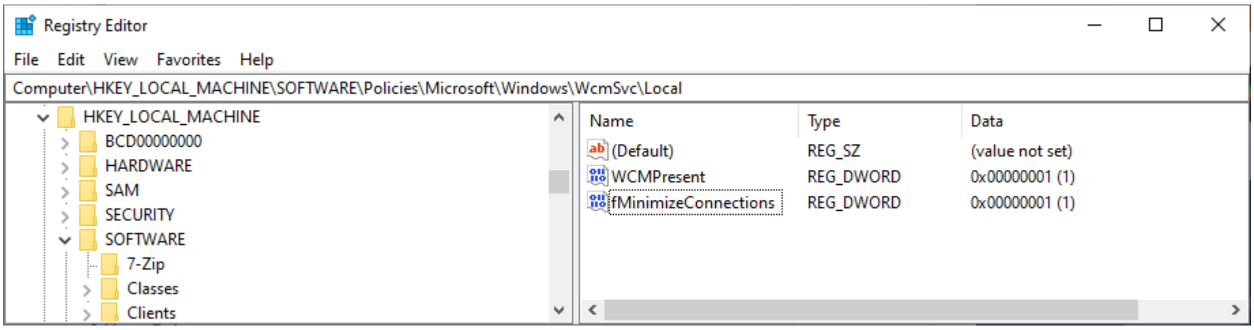


Figure 10: Windows registry

Warning: Please note that this is a change to the Windows system configuration and it will affect <u>any</u> cellular connection in the windows PC

For more details:

https://docs.microsoft.com/en-us/windows-hardware/drivers/mobilebroadband/understanding-and-configuring-windows-connection-manager

4. USE CASES AND COMPONENTS

4.1. Hardware Setup

1. Insert the Interface board with Telit Module on the EVB/EVK2 (step1)
2. Connect the power supply (step 2)
3. Connect the cellular antenna to the RF connector (step 3)
4. Connect the PC to the EVB using USB/Serial cable (step 4)
5. Insert the SIM card into the slot (step 5)
6. Power on the Telit Module pressing the "ON/OFF" button on the EVB/EVK2 for 6 seconds (step 6). Please refer to HW User guide [01] [02] [03] for more details about power on

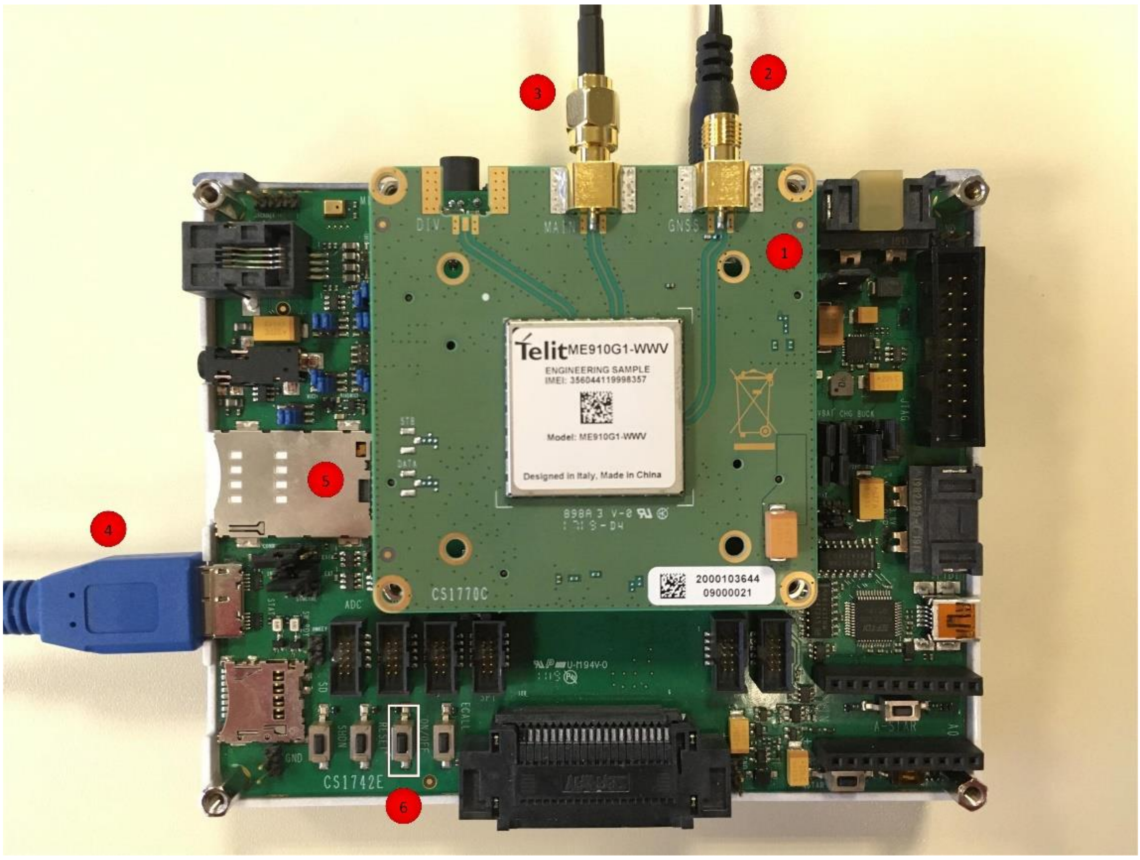


Figure 11: Joined elements.

4.2. LwM2M Registration to the OneEdge IoT Portal

The LwM2M client requires an active data connection in order to start the communication with the LwM2M server.

Figure 12 illustrates the basic actions to be performed in order to manually register the module to the GSM cellular network and then enable the LwM2M client to register to the OneEdge IoT Portal.

More details are explained below.

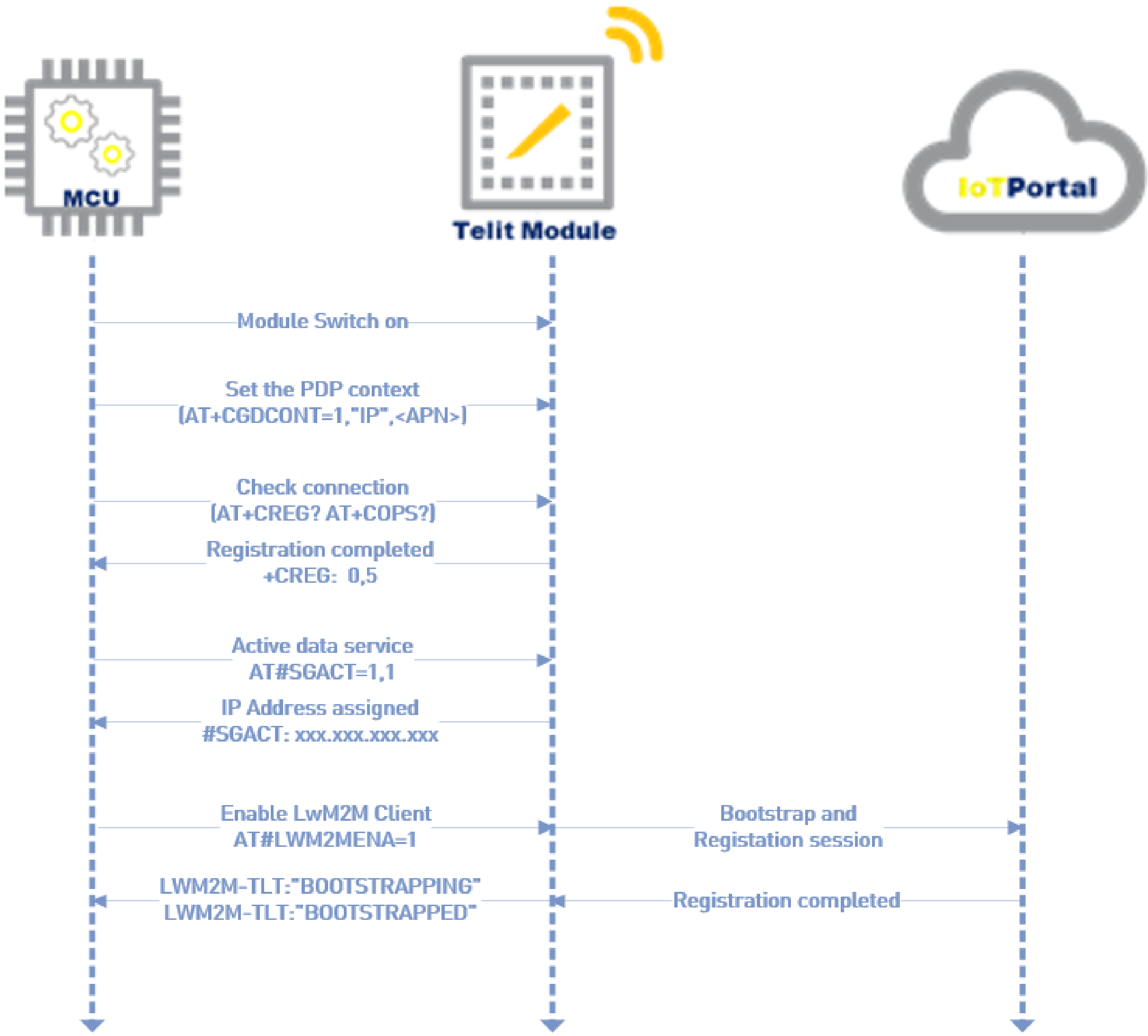


Figure 12: sequence of basic steps to perform GSM cellular registration and LwM2M connection

Since LwM2M is an application protocol at higher level over the UDP/IP stack, the logical pre-conditions to be satisfied are:

- cellular network registration;
- IP stack functionalities;
    - to be able to get an IP address;
    - to be able to successfully perform DNS queries;
    - to be able to send and receive ICMP data with a target server;
- LwM2M functionalities;
    - to be able to bootstrap and register to the LwM2M server;

The pre-conditions must be verified at least the first time that the EVB equipment is used (refer to paragraph 3.2).

The step-by-step procedure to verify the pre-conditions is described below.

<u>Cellular network registration:</u>

a) Depending on the cellular capabilities of the environment under test (GSM or CAT-M/NBIoT in case of MEx10, GSM or UMTS or LTE in case of LE910), the registration process could take many minutes.
Therefore, just for the first evaluation usage and to be sure that the environment is valid, the registration process can be speeded up to reduce the radio scanning time of the module.
To do that, force the modem to work only on a specific target RAT, according to the following table and then reboot the device (`AT#REBOOT`)

RAT technology	AT commands
GSM	AT+WS46=12
UMTS	AT+WS46=25
LTE	AT+WS46=28
CAT-M	AT+WS46=28 AT#WS46=0
NBIoT	AT+WS46=28 AT#WS46=1

Table 2: AT commands to force the modem to specific RAT only

b) The User must configure the suitable APN to be used by the LwM2M client (by default PDP 1 is used): `AT+CGDCONT=1,"IP",<APN>`

c) The User must send the `AT+CREG?` or `AT+CEREG?` commands ("CREG" for GSM/UMTS/LTE and "CEREG" for CAT-M/NBIoT) to check the cellular network registration status:

the network registration is completed when the following response is received: `+CREG/+CEREG: 0,1` or `0,5`.

d) The User must send the `AT+CGREG?` command to check if the module is properly registered the Packet Switching network:

the expected response is `+CGREG: 0,1` or `0,5`

e) The User must send the `AT+COPS?` command to verify the network operator which the device is camped to.

The response must report status and the name of the Network operator:

`+COPS: 0,0,"<MNO_name>",x`

Warning: If the modem is not able to register to the target cellular network, typical reasons can be: SIM not active, antenna disconnected, slow registration timing in case of NBIoT

IP stack functionalities:

When cellular registration is completed:

f) The User must activate data services and set up the internal IP stack with the command: `AT#SGACT=1,1;`

In case of success, the device receives a valid IP address from the Network Provider in the format: `#SGACT: xxx.xxx.xxx.xxx`

Tip: the Connection Agent can be very useful to performs all the previous cellular registration steps in a unique AT command.

For example:

```
AT#CASN=1
OK
#CAEV: SYS,4100,SET_PARAM_FROM_NVM_OK
#CAEV: ANT,3100,ANT_OK
#CAEV: NWK,1105,NET_STATUS_REG
#CAEV: DAT,2100,DATA_SERVICE_OK
#CAEV: DAT,2110,DATA_SERVICE_OK,10.7.35.205
```

---

g) The User must check if the modem is able to use the DNS service through the `AT#QDNS="bs.telit.io"` command;

if the DNS query is successful, an IP address is reported on the second parameter of the response, otherwise an empty string in case of failure

Successful example: `#QDNS: "bs.telit.io","54.93.92.219"`

Failed example: `#QDNS: "bs.telit.io",""`

Warning: If the modem is not able to resolve the server address, typical reasons can be: no DNS server has been assigned to the modem (sometimes in NBIoT networks), the DNS server is not able to provide any IP address. Technical support team can analyse the logs and address the right cause and solution.

h) The User must check if the modem is able to use the IP stack through the `AT#PING="www.telit.com"` command;

if the ping is successful, four rows are reported and last 2 parameters of each one should be different than "600,255":

Successful example: `#PING: 01,"35.202.235.194",xxx,yyy`

Failed example: `#PING: 01,"35.202.235.194",600,255`

Warning: If the modem is not able to ping the server address, typical reasons can be: latency very high (often in NBIoT), firewall or communication blocked by MNO. Technical support team can analyse the logs and address the right cause and solution.

For more details, please refer to the document "2G 3G 4G Registration Process 80000NT11696A".

Furthermore, most of the steps described above could be performed using the Connection Agent. The Connection Agent functionalities automatically connect the module to the network, set and enable the PDP context (refer to "Connection Agent User Guide" [11] for more details).

<u>LwM2M functionalities:</u>

When also IP functionality check is completed:

i) The User must enable the additional DM URC Service through the `AT#LWM2MW=0,33211,0,0,1,1` command;

j) The User must activate the LwM2M client through the `AT#LWM2MENA=1` command:

   If the LwM2M client is not bootstrapped yet, the Bootstrap procedure is performed, so the following Bootstrap URC messages are received:

   `LWM2M-TLT:"BOOTSTRAPPING",SSID=0,"coaps://bs.telit.io"`

   `LWM2M-TLT:"BOOTSTRAPPED",SSID=0,"coaps://bs.telit.io"`

   In any case, the registration procedure is performed, so the following additional URC are received

   `LWM2M-TLT:"REGISTERING",SSID=99,"coaps://api-dev.devicewise.com"`

   `LWM2M-TLT:"REGISTERED",SSID=99,"coaps://api-dev.devicewise.com"`

k) The User must send again the `AT#LWM2MW=0,33211,0,0,1,1` command to let the LwM2M client to save it in NV memory;

The connection between the LwM2M client and the LwM2M Server is now up and running.

Tip: the set values through the AT Command `AT#LWM2MW=0,33211,0,0,1,1` are persistent on power cycles because they are saved into NV memory. They are reset only across the bootstrap procedure.

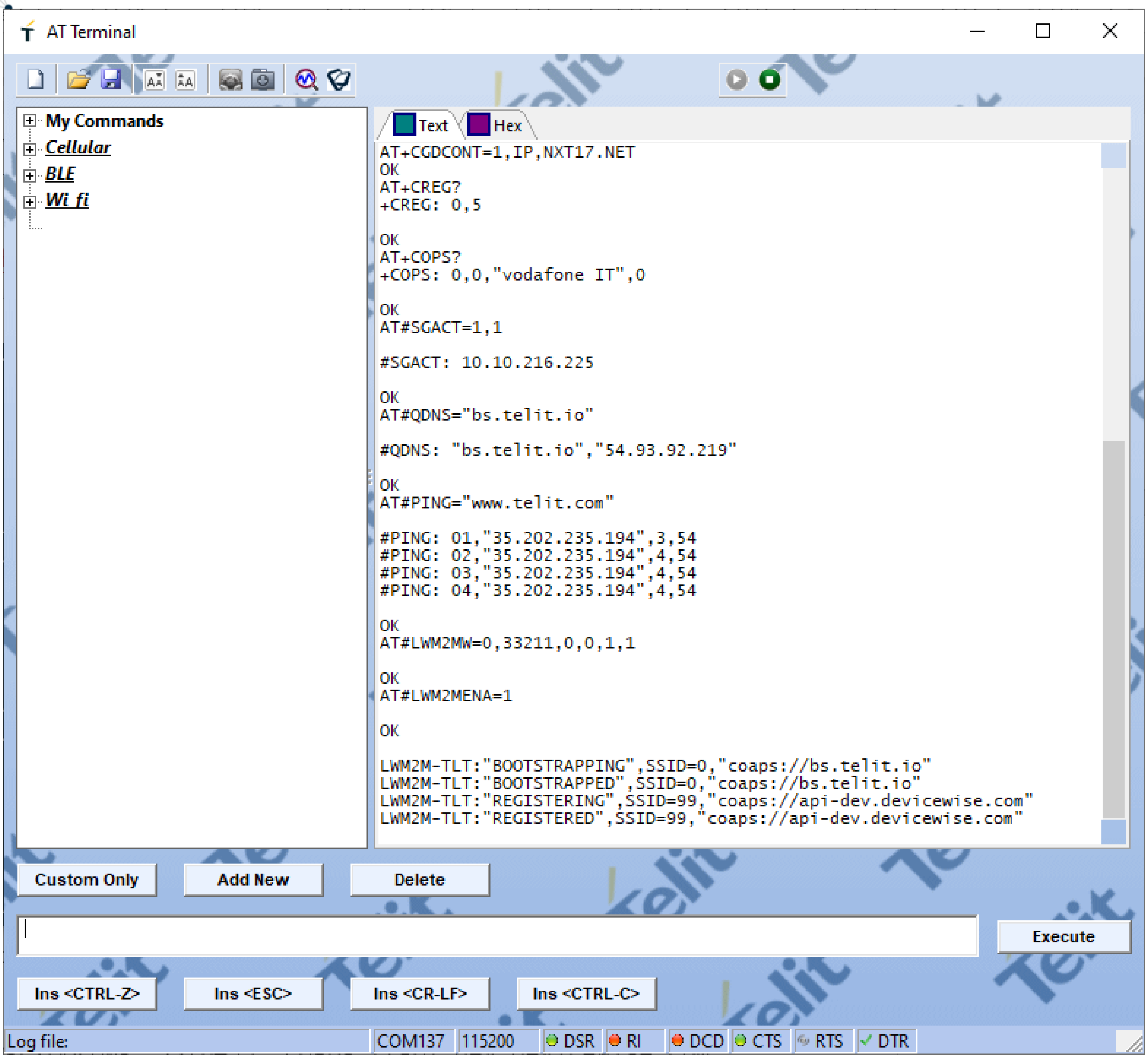


Figure 13: sequence of commands for cellular GSM registration, IP functionality check and LwM2M registration

At this point, as soon as the bootstrap phase is finished and despite the URC related to the connection with the DM Server are not visible, the User can check on the OneEdge IoT Portal whether the LwM2M registration has been completed correctly:

On OneEdge IoT Portal, on Thing > **Overview** tab, the "Last seen" field reports:

- Green clip: Lifetime is not expired – thing is actually connected;
- Elapsed time since the last Registration Update.

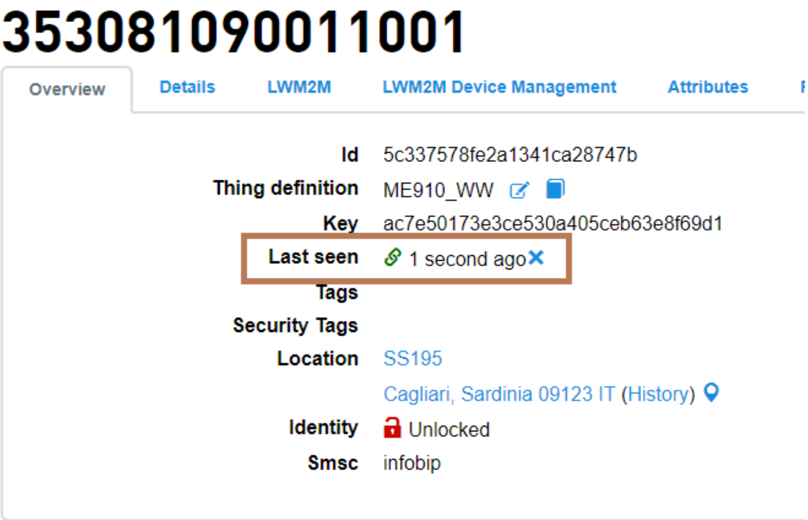


Figure 14: Thing Example

4.3. Handle the LwM2M Objects

According to the previous steps, the following preconditions are satisfied:

- Bootstrap & Registration already performed;
- LwM2M client correctly connected.

4.3.1. Four basic User Scenarios

Related to the data exchange, OneEdge allows to perform four different basic operations:

- **"Reading" scenario**: the host application can access to the information available in the LwM2M client through AT Commands; the User can read the same information from the OneEdge IoT Portal;
- **"Remote setting" scenario - Monitoring**: the User changes parameters of the remote device through the OneEdge IoT Portal and the host application is informed about it;
- **"Remote setting" scenario - Exec**: Exec operation is sent from the OneEdge IoT Portal to the host application;
- **"Metering" scenario**: the User can collect remote sensor data and analyzes its trend on the OneEdge IoT Portal.

OneEdge also provides three ways to access/modify the Resources stored in the Object Repository within the LwM2M client: the AT commands on the host side, OneEdge IoT Portal interface and the AppZone APIs on the modem side.

4.3.1.1. “Reading” scenario

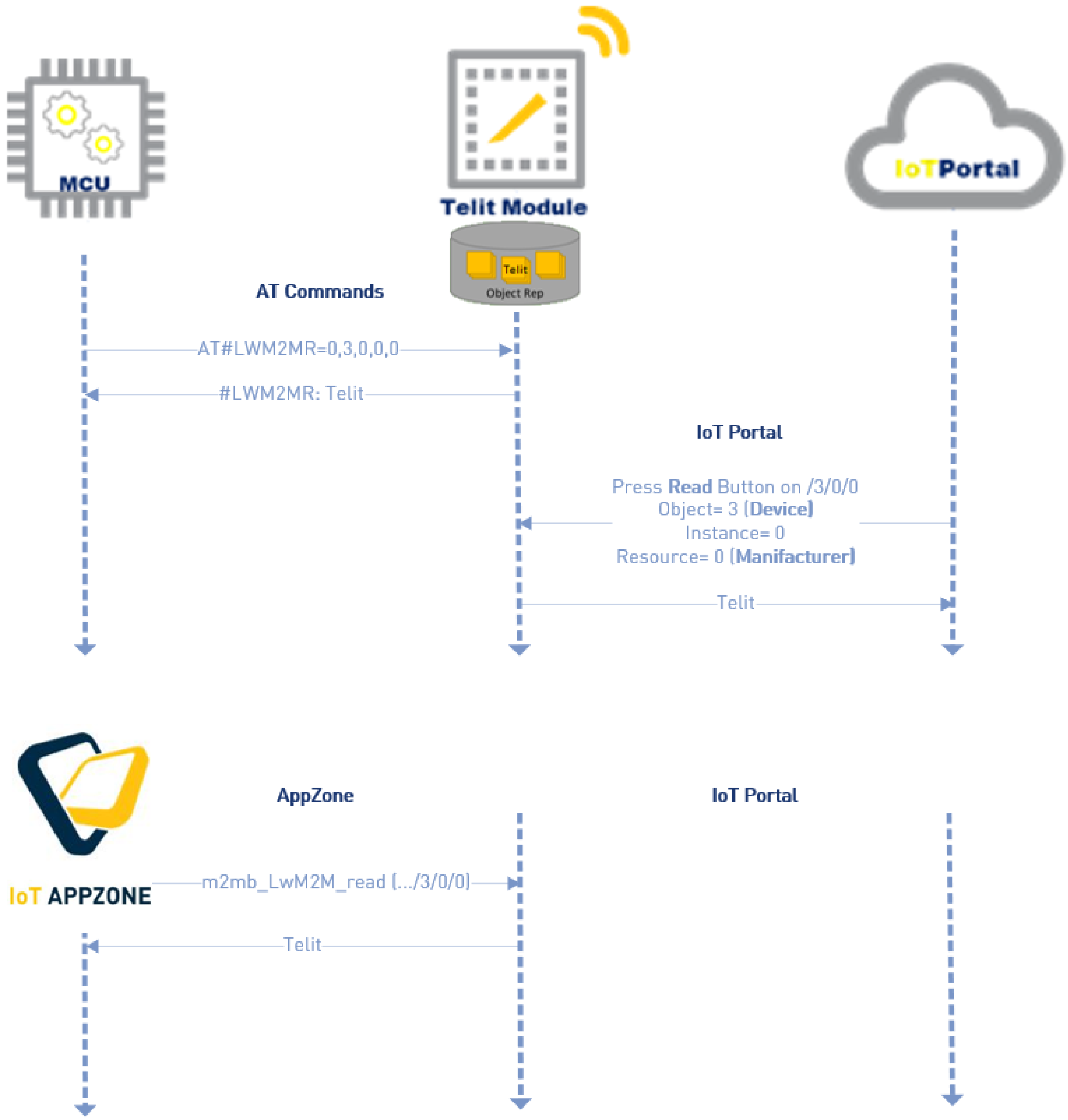


Figure 15: Read a Resource

4.3.1.1.1. AT Commands

To read an object resource from the External MCU (host), the User shall send the AT#LWM2MR command with the specific URI values.

The example below is reading the resource value /3/0/0 (Device Manufacturer of the Device object) with is the string “Telit”

```
AT#LWM2MR=0,3,0,0,0
#LWM2MR: Telit
OK
```

4.3.1.1.2. OneEdge IoT Portal Interface

From the OneEdge IoT Portal interface:

- Go to **Things** > Select device > **LwM2M** tab > **Object browser** button.
- Select the Object > Resource you want to read and press the **Read** button as shown in Figure 16.
- The tip on the **Read** button will report a short log (useful mainly in case of error). The read value will be displayed in the column on the right.

Figure 16: Read a Resource from the OneEdge IoT Portal

Note: If you want to read all the Resources within an object, from the OneEdge IoT Portal, click on the **Read** button associated to the Object.

4.3.1.1.3. AppZone

To read a resource via AppZone environment:

- The LwM2M client must be already set and enabled in order to correctly run the read command (using *m2mb_LwM2M_init* and *m2mb_LwM2M_enable*).
- The User must create an AppZone application that implements the main logic and call the corresponding API *m2mb_LwM2M_read*.

Note: A sample application related to some LwM2M features is available on:

https://github.com/telit/IoT-AppZone-SampleApps/tree/master/ME910C1-ML865C1-NE910C1/AppZoneSampleApps-USB0/LWM2M

4.3.1.2. "Remote Setting" Scenario - Monitoring

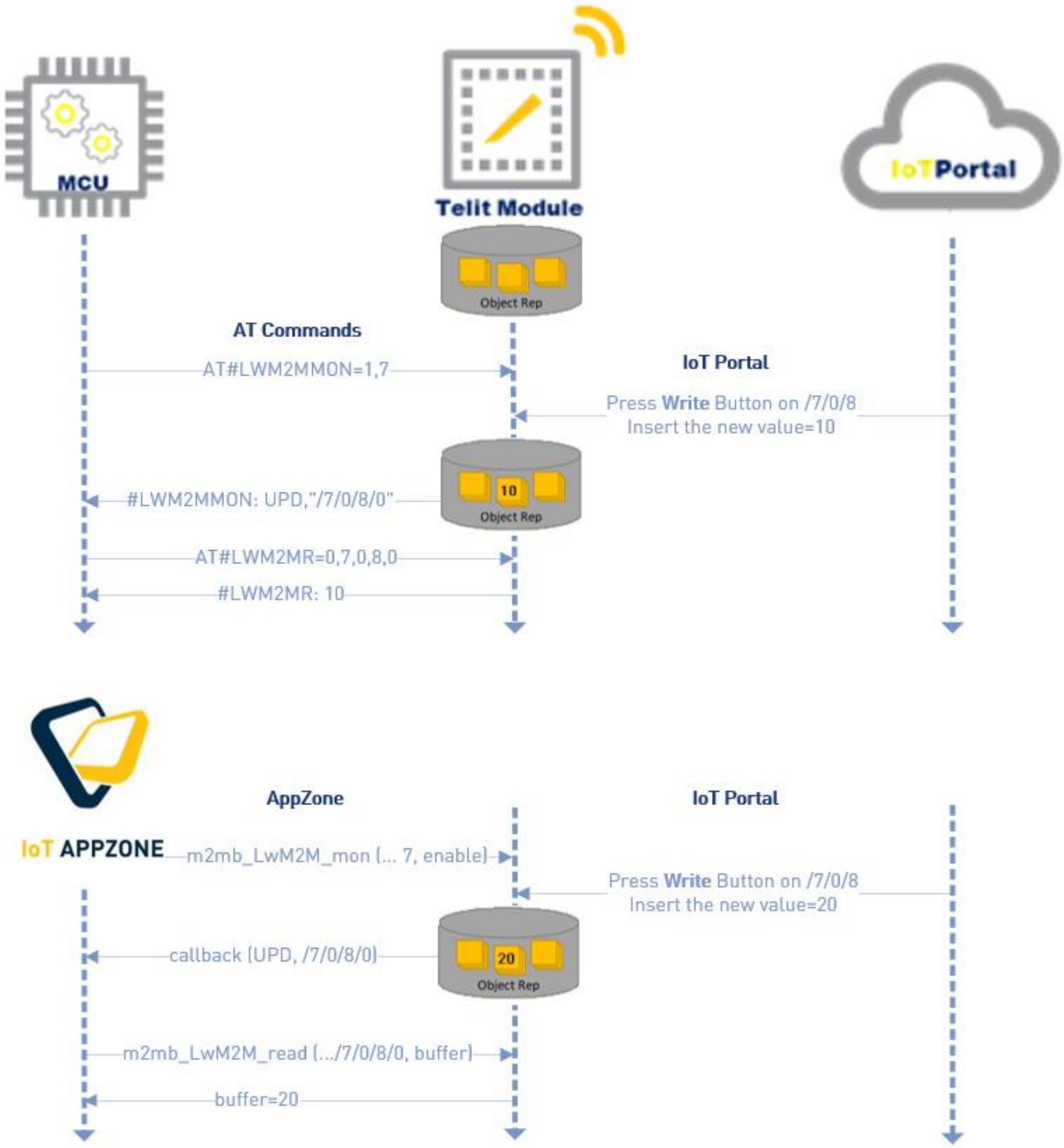


Figure 17: Write a Resource

4.3.1.2.1. AT Commands

The Monitoring feature is very useful to inform a host application when a resource has been changed.

The AT command to enable this functionality is AT#LWM2MMON, it allows to obtain real-time information (in the #LWM2MMON: UPD format) on the host application, as a result of any write operation.

```
AT#LWM2MMON=1,7
OK
```

Here the User writes the value "10" to the resource /7/0/8 on the OneEdge IoT Portal User interface, as described in the next paragraph.

The host application receives the following message:

```
#LWM2MMON: UPD,"/7/0/8/0"
```

But, as the URC does not contain the value of the resource, the host application can achieve it through a "Read" operation.

```
AT#LWM2MR=0,7,0,8,0
#LWM2MR: 10
OK
```

Tip: Here the host can take some actions according to the read value, for example change the value of a GPIO to switch on a LED, open/close a relay, start an engine, etc.

4.3.1.2.2. OneEdge IoT Portal Interface

From the OneEdge IoT Portal interface:

- Go to **Things** > Select device > **LwM2M** tab > **Object browser** button.
- Select the desired Object > Resource > press the **Write** button.
- A pop-up with the specific field to be configured will appear. Enter a valid value (according to the field type) and press **Update**.

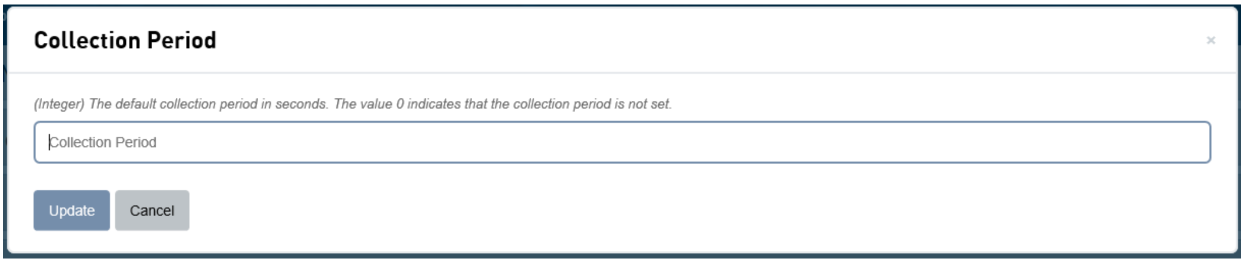


Figure 18: Object Browser

- If the write action is performed correctly, the **Write** button will be colored green. Otherwise it will be red: in this case, the cause of the error will be shown by positioning the mouse cursor over the red icon.

  `#LWM2MMON: UPD,"/7/0/8/0"` is received on the serial port to inform the host that the 7/0/8 resource has been modified.

4.3.1.2.3. AppZone

The same actions performed by the AT commands could be replaced by the following AppZone functions.

The corresponding API for AT#LWM2MMON is *m2mb_LwM2M_mon*.

The URC "#LWM2MMON: UPD, ..." is detected by the LWM2M AppZone callback.

The corresponding API for AT#LWM2MR is *m2mb_LwM2M_read*.

4.3.1.3. "Remote Setting" Scenario - Exec

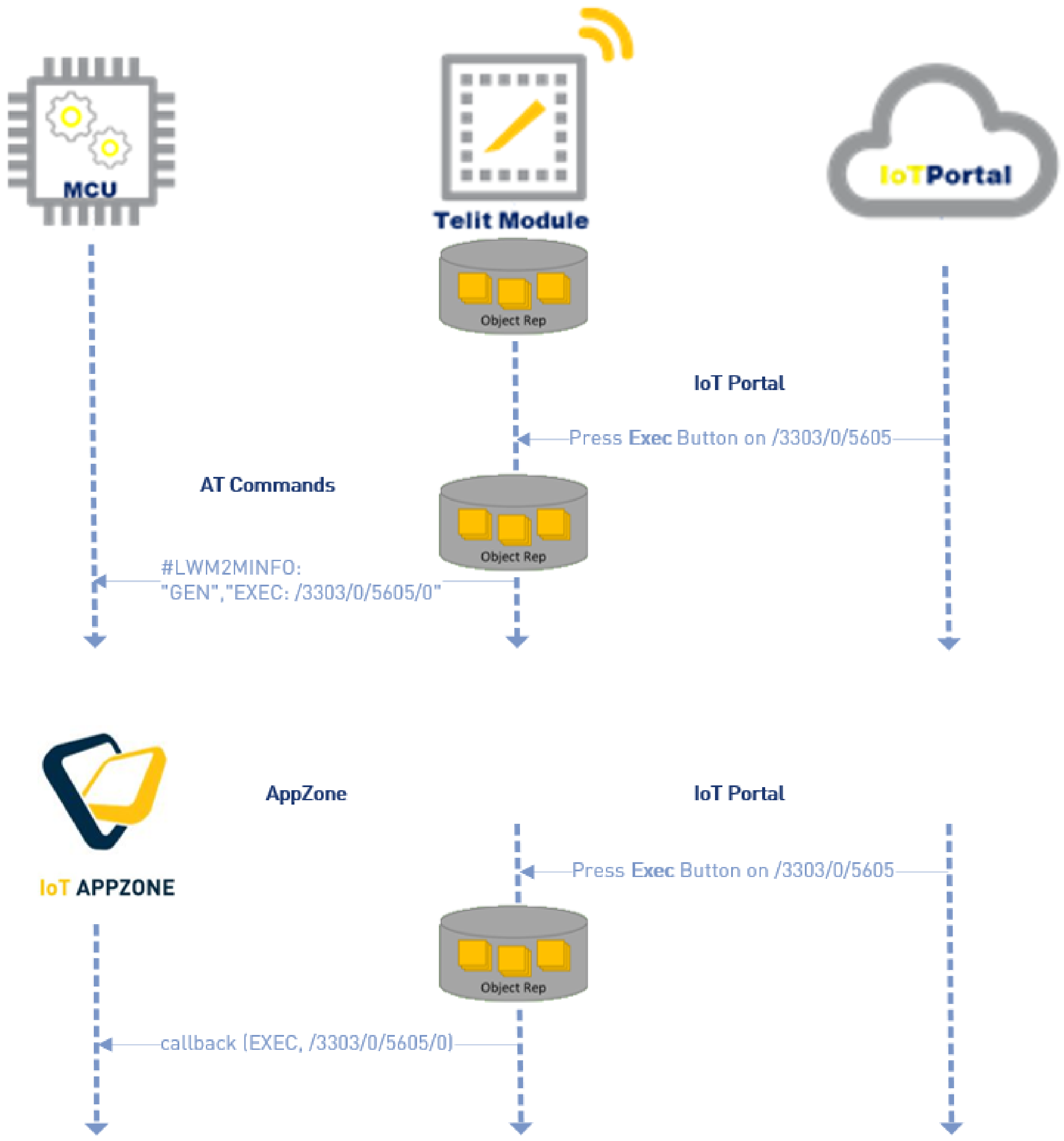


Figure 19: Exec a Resource

4.3.1.3.1. AT Commands

There are some resources defined as "Executable".

The starting point, in this case, is only the OneEdge IoT Portal.

When a resource like that is triggered, the host application receives the following URC:

```
#LWM2MINFO: "GEN","EXEC: /objID/instID/resID/resInstID"
```

Here the host can take some actions according to the executed resource, for example change the value of a GPIO to switch on a LED, open/close a relay, start an engine, etc

4.3.1.3.2. OneEdge IoT Portal Interface

From the OneEdge IoT Portal interface:

- Go to **Things** > Select device > **LwM2M** tab > **Object browser** button.
- Select the desired Object > Resource > press the **Exec** button.

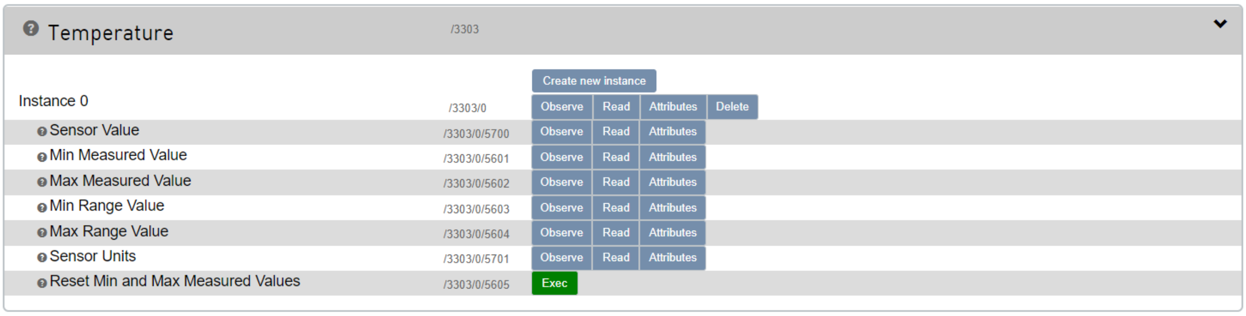


Figure 20: Object Browser

- If the action is performed correctly, the **Exec** button will be colored green. Otherwise it will be red: in this case, the cause of the error will be shown by positioning the mouse cursor over the red icon.

  `#LWM2MINFO: "GEN","EXEC: /3303/0/5605/0"` is received on the serial port to inform the host that the 3303/0/5605 resource has been executed.

4.3.1.3.3. AppZone

The Exec message "#LWM2MINFO: "GEN","EXEC..." can be managed by AppZone code through the LWM2M callback.

4.3.1.4. "Metering" Scenario

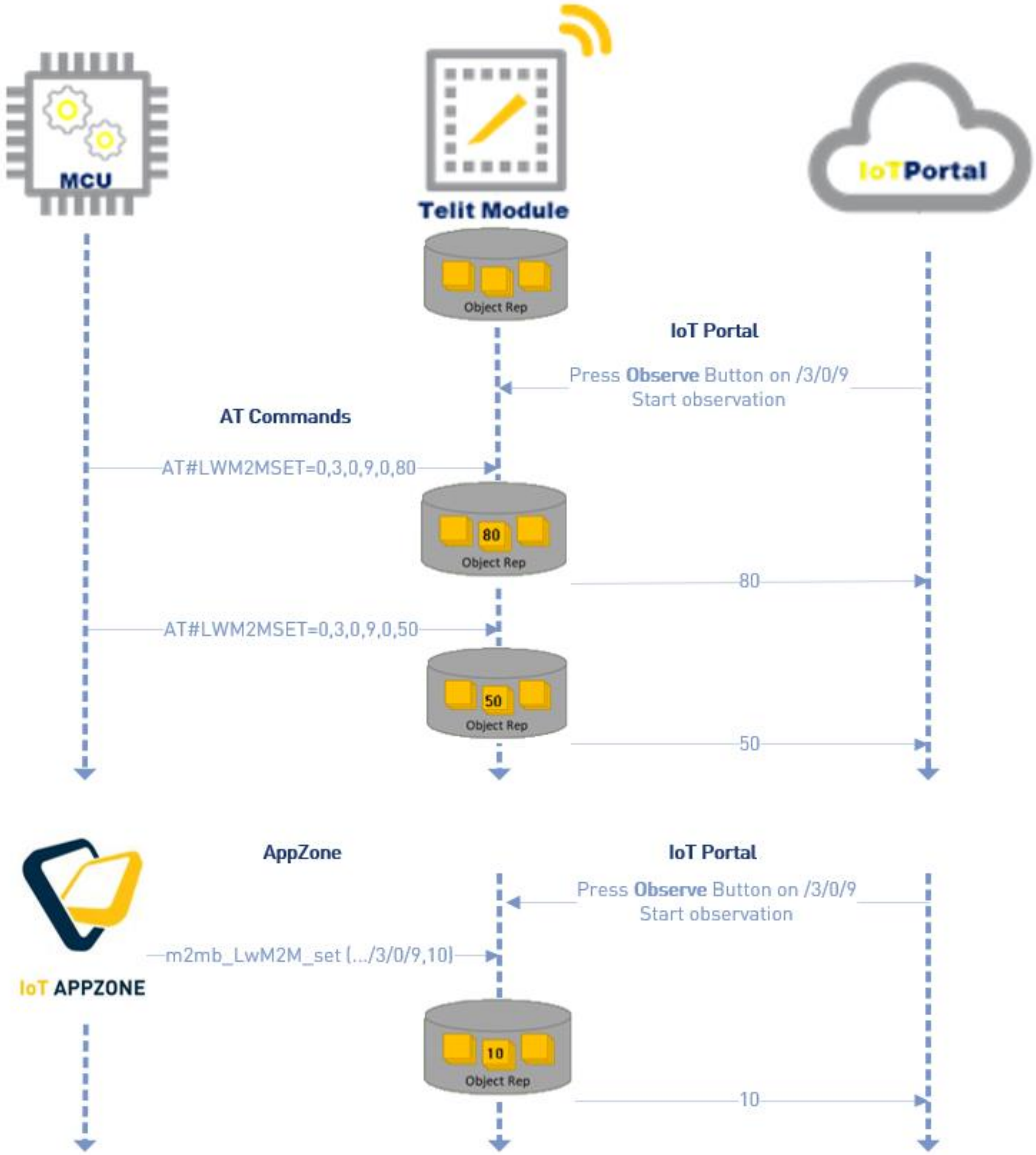


Figure 21: Observe a Resource

The "Observe" operation is associated to a specific resource and allows to monitor the value of that resource by sending a notification to the OneEdge IoT Portal whenever that value changes.

4.3.1.4.1. AT Commands

Precondition: the User sets the observation on the OneEdge IoT Portal side for the desired resource.

The host application writes a value in a resource through the AT#LWM2MW (for Read/Write resources) and AT#LWM2MSET (for Read Only resources) commands, and according to the observation, the LwM2M client sends a notification with the value to the OneEdge IoT Portal.

The example below, showed in Figure 22, is related to the setting of the value "80" to resource /3/0/9 (Battery Level of the Device object).

```
AT#LWM2MSET=0,3,0,9,0,80
OK
```

Furthermore, the host application can be informed about the delivery status of the set value.

The following AT commands can be used:

```
AT#LWM2MNFYACKURI=0,1,3,0,9
AT#LWM2MNFYACKENA=0,1
```

In case of successful notify, the host receives this URC message:

```
#LWM2MNFYACK: 0,99,"/3/0/9","ACK"
```

Otherwise:

```
#LWM2MNFYACK: 0,99,"/3/0/9","NACK"
```

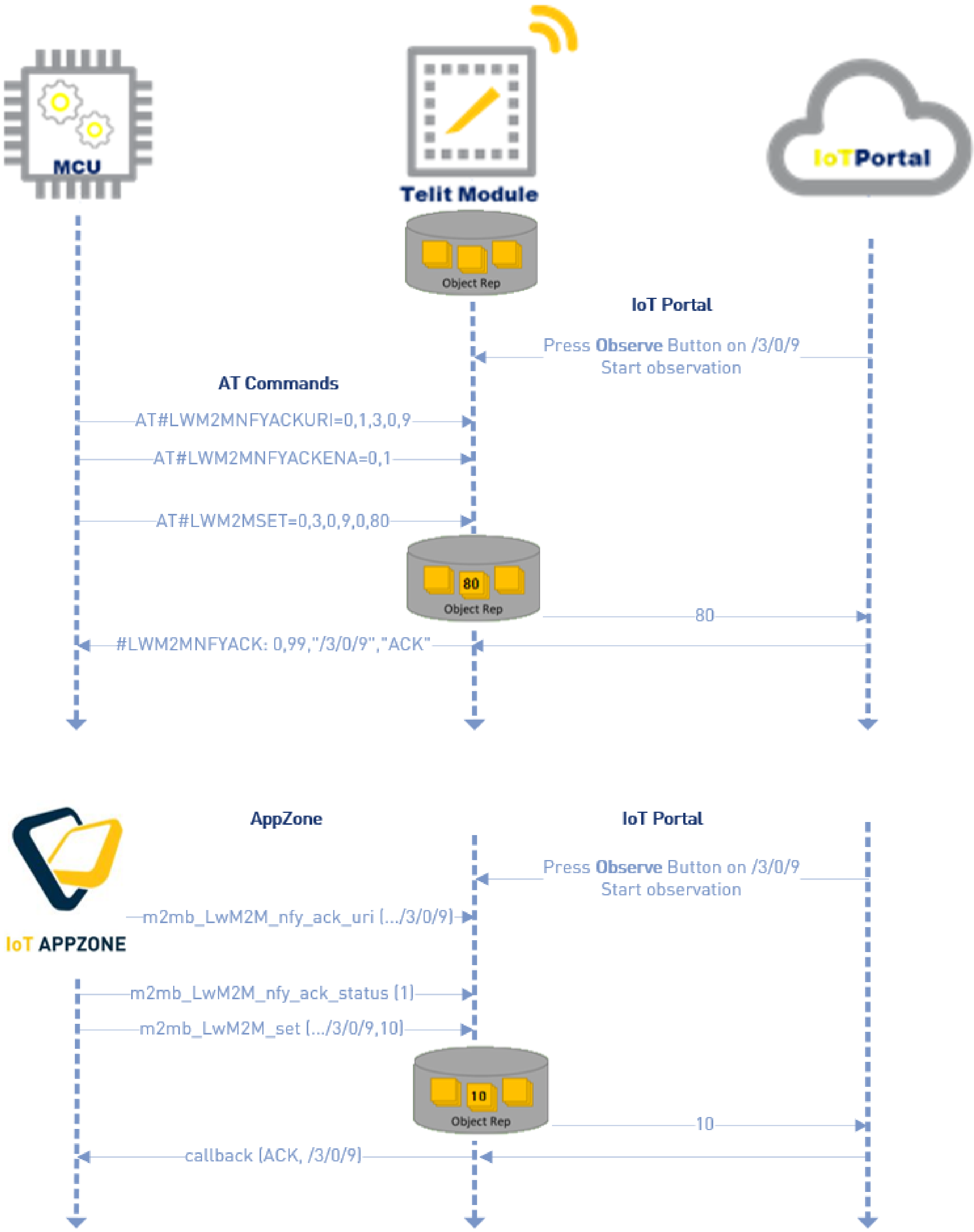


Figure 22: ACK URC Reporting

4.3.1.4.2. OneEdge IoT Portal Interface

To enable observation for a resource:

- Go to **Things** > Select device > **LwM2M** tab > **Object browser** button.
- Select the desired Object > Resource to be observed and press the **Observe** button

Figure 23: Observe activation

By default, the notification will be sent to the OneEdge IoT Portal whenever the resource value changes without any restrictions.

4.3.1.4.3. AppZone

The same actions performed by the AT commands could be replaced by the following AppZone functions.

The corresponding API for AT#LWM2MSET is *m2mb_LwM2M_set*.

The corresponding API for AT#LWM2MNFYACKURI and AT#LWM2MNFYACKENA are *m2mb_lwm2m_nfy_ack_uri* and *m2mb_lwm2m_nfy_ack_status*

The URC "#LWM2MNFYACK:…, ACK" is managed by the LWM2M AppZone callback.

4.3.1.4.4. Attributes Options

Notification events can be customized by changing the "attributes" parameters.

To edit the attributes of a specific resource, select the Object > Resource under observation and press the **Attributes** button.

Figure 24 shows the parameters that could be changed:

- **Minimum period** and **Maximum period**: values expressed in seconds, they define a time range within which the client sends a notification in case the resource value changes. The time count starts from the last change of the value of the resource.

In case the resource value changes before the minimum period the notification will be sent exactly to the Minimum period. When the elapsed time reaches the Maximum period, a notification will be sent even if the value is not changed.

- **Step**: the client sends a notification to the server if the difference between the old and the new value is equal to or greater than the step value.
- **Greater than** and **Less than**: a notification is sent only if the new value crosses one of these thresholds.

Warning: "Step", "Greater than" or "Less than" are considered only when they fall within the time range defined by "Minimum Period" and "Maximum Period".

Note: The five attributes can be set separately or in different combinations.

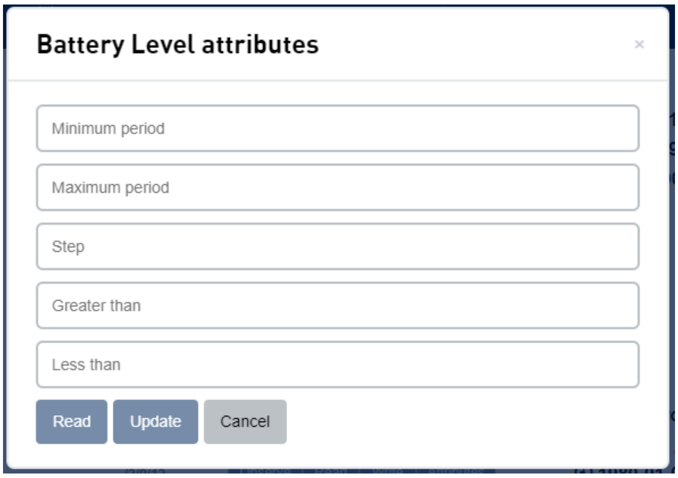


Figure 24: Attribute's example

Both observations and attributes can be enabled and set only from the OneEdge IoT Portal interface (no way to handle them from the host side).

4.4. Handle the Device from the OneEdge IoT Portal Interface

This section describes how to handle the devices from the OneEdge IoT Portal interface. No actions are required from the device.

4.4.1. List the Devices

By default, the OneEdge IoT Portal interface provides the list of all Things associated with the specific Organization.

The User can find the list of his Things by pressing the **Things** tab > **Table** tab

By default, each device is identified by a specific and unique name.

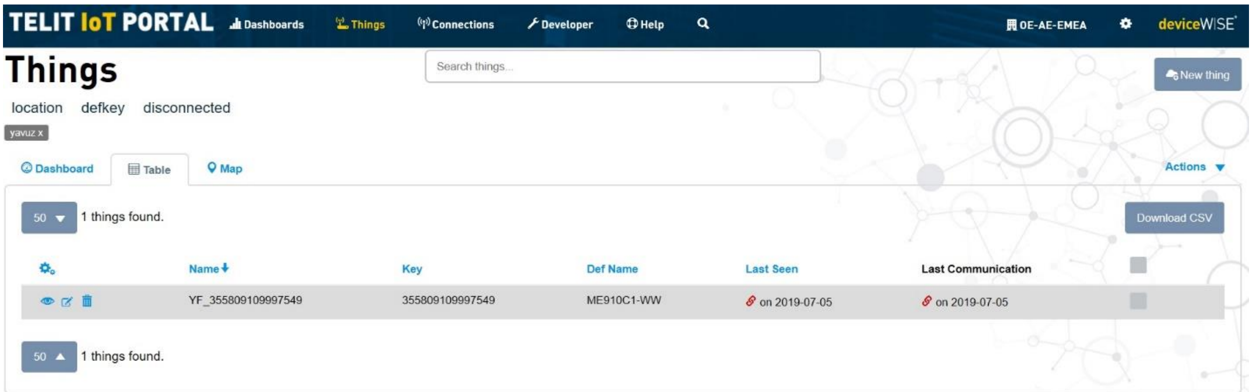


Figure 25: Things page

4.4.2. Thing Details

The User can browse details of the thing by clicking **Things** tab > **Table** tab > **View** button .

The **LwM2M** tab appears showing the device identity, Remote address (IP address) of the device, Protocol, Binding and Lifetime information.

The **LwM2M** tab includes:

- **Object browser**: The Object browser lists all the objects/instances/resources supported by the thing.
- **Set profile**: The Set profile allows to change the Device Profile of the LwM2M thing.
- **Queue history**: all stacked up requests are listed in the Queue history.
- **Wakeup**: It sends an SMS to wake up the device.

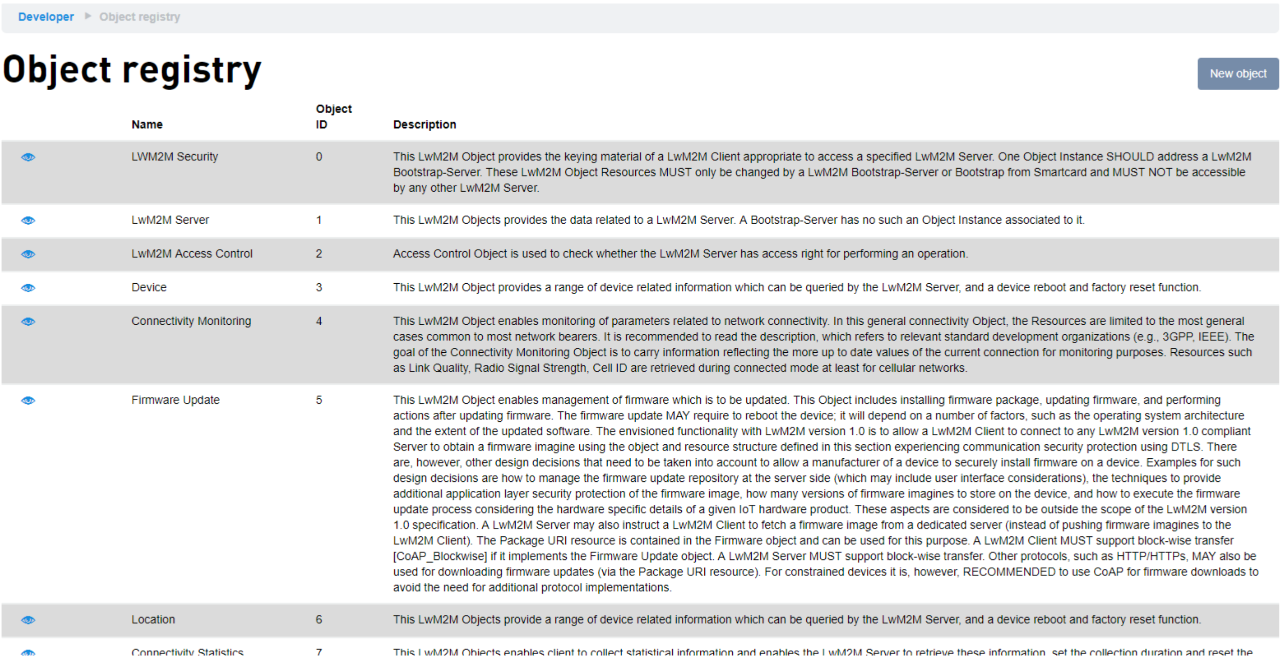
4.4.3. Create/Delete a Custom Objects

Each Telit device is provided with a set of standard objects, as per LwM2M OMA Specifications, and also with some Telit objects.

There is no way to modify these objects: they are mandatory to the proper functioning of the LwM2M client and also provide some additional features.

In case the User needs to add a new Object (not stored in the module), it is possible to add it manually.

The object structure is defined by XML file. In case the Object to add belongs to the standards ones, the XML structure can be retrieved directly in the **Object registry** of the OneEdge IoT Portal.



Developer ▸ Object registry

Object registry

New object

Name	Object ID	Description
LWM2M Security	0	This LwM2M Object provides the keying material of a LwM2M Client appropriate to access a specified LwM2M Server. One Object Instance SHOULD address a LwM2M Bootstrap-Server. These LwM2M Object Resources MUST only be changed by a LwM2M Bootstrap-Server or Bootstrap from Smartcard and MUST NOT be accessible by any other LwM2M Server.
LwM2M Server	1	This LwM2M Objects provides the data related to a LwM2M Server. A Bootstrap-Server has no such an Object Instance associated to it.
LwM2M Access Control	2	Access Control Object is used to check whether the LwM2M Server has access right for performing an operation.
Device	3	This LwM2M Object provides a range of device related information which can be queried by the LwM2M Server, and a device reboot and factory reset function.
Connectivity Monitoring	4	This LwM2M Object enables monitoring of parameters related to network connectivity. In this general connectivity Object, the Resources are limited to the most general cases common to most network bearers. It is recommended to read the description, which refers to relevant standard development organizations (e.g., 3GPP, IEEE). The goal of the Connectivity Monitoring Object is to carry information reflecting the more up to date values of the current connection for monitoring purposes. Resources such as Link Quality, Radio Signal Strength, Cell ID are retrieved during connected mode at least for cellular networks.
Firmware Update	5	This LwM2M Object enables management of firmware which is to be updated. This Object includes installing firmware package, updating firmware, and performing actions after updating firmware. The firmware update MAY require to reboot the device; it will depend on a number of factors, such as the operating system architecture and the extent of the updated software. The envisioned functionality with LwM2M version 1.0 is to allow a LwM2M Client to connect to any LwM2M version 1.0 compliant Server to obtain a firmware imagine using the object and resource structure defined in this section experiencing communication security protection using DTLS. There are, however, other design decisions that need to be taken into account to allow a manufacturer of a device to securely install firmware on a device. Examples for such design decisions are how to manage the firmware update repository at the server side (which may include user interface considerations), the techniques to provide additional application layer security protection of the firmware image, how many versions of firmware imagines to store on the device, and how to execute the firmware update process considering the hardware specific details of a given IoT hardware product. These aspects are considered to be outside the scope of the LwM2M version 1.0 specification. A LwM2M Server may also instruct a LwM2M Client to fetch a firmware image from a dedicated server (instead of pushing firmware imagines to the LwM2M Client). The Package URI resource is contained in the Firmware object and can be used for this purpose. A LwM2M Client MUST support block-wise transfer [CoAP_Blockwise] if it implements the Firmware Update object. A LwM2M Server MUST support block-wise transfer. Other protocols, such as HTTP/HTTPS, MAY also be used for downloading firmware updates (via the Package URI resource). For constrained devices it is, however, RECOMMENDED to use CoAP for firmware downloads to avoid the need for additional protocol implementations.
Location	6	This LwM2M Objects provide a range of device related information which can be queried by the LwM2M Server, and a device reboot and factory reset function.
Connectivity Statistics	7	This LwM2M Objects enables client to collect statistical information and enables the LwM2M Server to retrieve these information, set the collection duration and reset the

Figure 26: Object registry

Warning: Please be aware that the Object ID 0 "LWM2M Security" can not be modified, read or write under any circumstances.

- Select **Developer** tab > **Object registry**
- Select the desired Object then click on "**Download XML**" button
- The definition file will be downloaded

At this point the new Object must be uploaded on the module. To upload the XML file in module's memory you need to use an AT terminal program that permits also to transfer files.

Please be sure that, in case of big XML files, the HW flow control is enabled on both the serial terminal tool and the modem side through the AT&K3 command.

The settings on TATC to enable the HW flow control is showed in Figure 27:

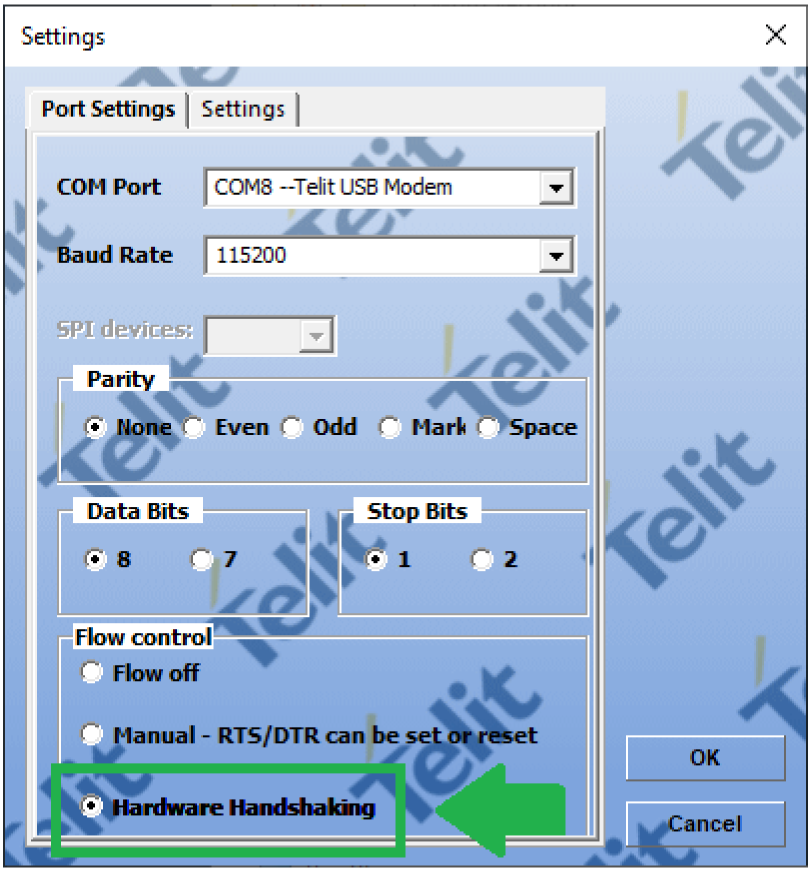


Figure 27: Flow control set as "Hardware Handshaking"

To do this use the following AT command:

a. AT#M2MWRITE=/XML/<xml_file_name>,<file_size>

After that the User needs to use the tool to transfer the .xml file, the **Transfer Data** icon can be pressed to do this.

b. Click on the Transfer Data icon and select the <xml_file_name>.xml from the PC

c. The tool will transfer the file and then if all is ok, the module will report with OK on the output list.

d. Reboot the module through `AT#REBOOT`;

e. Start the LwM2M Agent with `AT#LWM2MENA=1`;

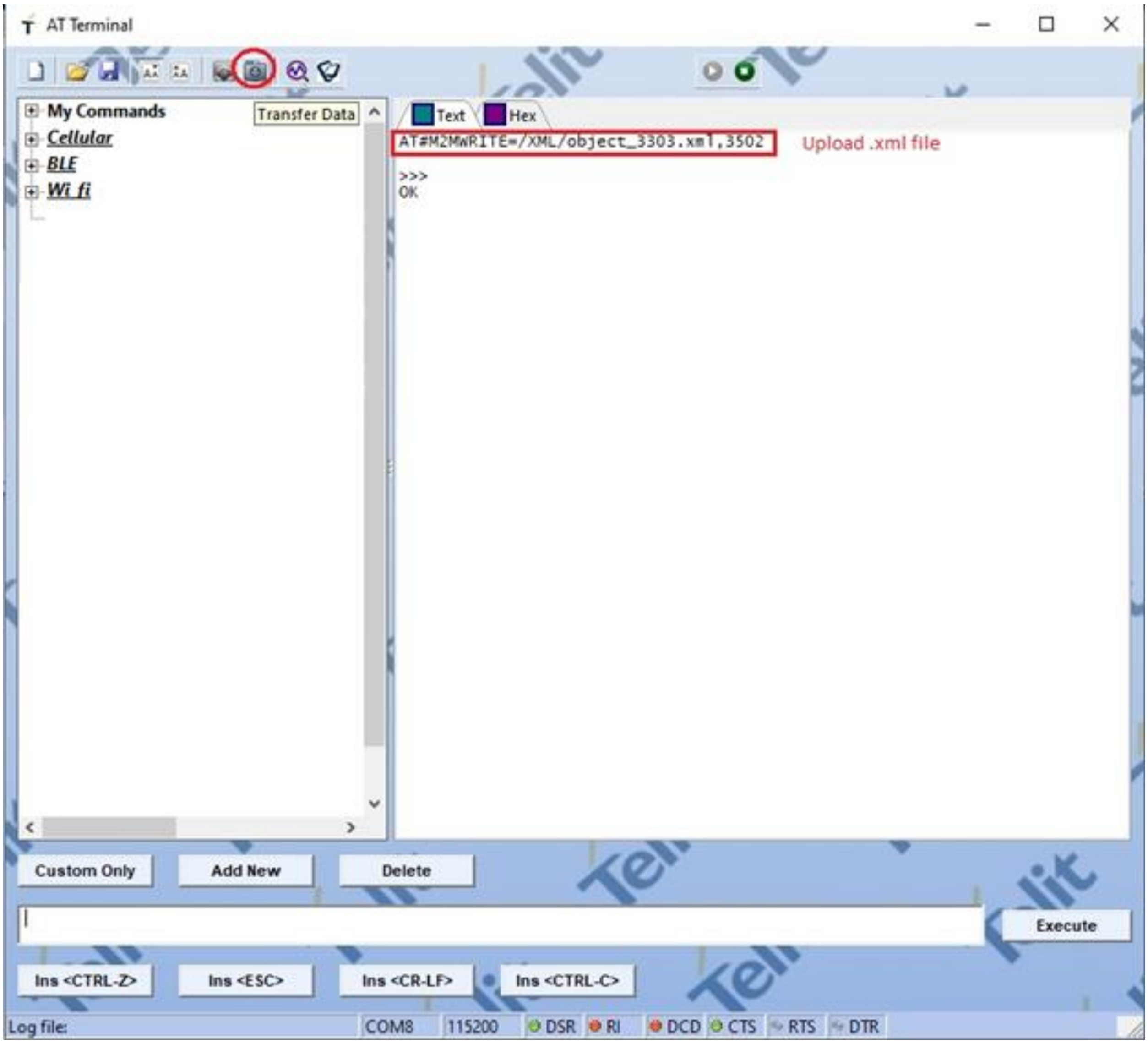


Figure 28: Object 3303 "Temperature" upload on module.

The file is now installed in the module and the new object is ready to be instantiated.

f. Create the Instance using the `AT#LWM2MNEWINST` command;

The syntax is the following:
`AT#LWM2MNEWINST=<agentInstance>,<objectID>,<objectInstanceID>`

Alternatively to step f, the same operation of Instance creation could be done via OneEdge IoT Portal, as showed on Figure 29:

g. Go to **Things** > Select your device > **LwM2M** tab > **Object browser** button.

h. Search the created Object and press **Create New Instance** Button

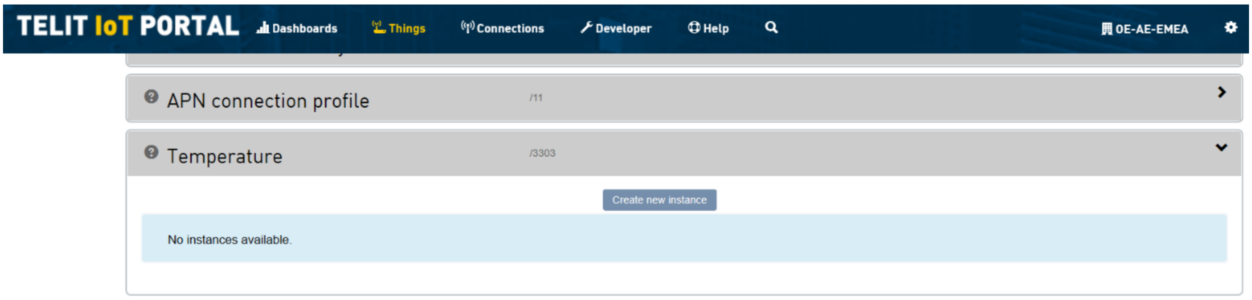


Figure 29: Object browser of thing

In case the Object is not standard and also not present in the Object registry, the User must create the XML file following the XML Schema defined by OMA, then follow the same procedure to upload it into the module.

Warning: The XML file is supposed to be well-formed when loaded in the module. The validation of the XML definition is up to the User. In case the XML format is not correct the LwM2M client will not be able to correctly load it. We strongly recommend to use the official OMA LwM2M object editor: https://devtoolkit.openmobilealliance.org/OEditor/default.aspx

---

In case the Object is not standard, the User must upload the XML definition in the OneEdge IoT Portal too:

- Click in **Developer** tab > **Object Registry**
- Click New Object
- Paste the XML definition

To delete a custom Object, instead, simply remove the XML file from the module. Use:

```
AT#M2MDEL=/XML/<xml_file_name>
```

5. SELF DIAGNOSTIC

The following information may provide solutions to typical problems before to involve OneEdge technical support team.

Problem	Cause	Action
The module does not complete the network registration	1. The SIM card is not inserted 2. The SIM card is not enabled 3. Cellular Antenna is not connected 4. No Network coverage 5. The APN has not been set correctly 6. The network adapter has not been disabled	1. Insert the SIM card 2. Contact SIM provider 3. Connect the antenna as described in the section"4.1. Hardware setup" 4. Verify the service coverage (refer to "4.2. Data connection to the OneEdge IoT Portal" paragraph) 5. Verify the network operator information and perform the step described in the section "4.2. Data connection to the OneEdge IoT Portal" 6. Follow the instructions on chapter "3.3.1.Disable network adapter"
The client fails the bootstrap for three times	1. Empty **Device Profile** on thing configuration	1. On OneEdge IoT Portal, check the **LWM2M** tab of the thing view. If the field is empty, please contact the support providing this information
The host application does not receive the URC registering/registered messages	1. The AT command AT#LWM2MW=0,33211,0,0,1,1 was never sent	1. Run AT#LWM2MW=0,33211,0,0,1,1
The host application fails to write a value on a resource	1. `AT#LWM2MSET` used for "Read/Write" resources 2. `AT#LWM2MW` used for "Read Only" resources	Check the object properties on **Object registry** of **Developer** section: 1. Use `AT#LWM2MSET` for "Read only" resources 2. Use `AT#LWM2MW` for "Read/Write" resources
The OneEdge IoT portal returns error and pressed button appears red during object browser actions (Read, Write, Observe, Attributes)	1. The LWM2M message sometimes is lost 2. The portal always fails to deliver the LWM2M message to the device	1. Try again 2. On portal side, set lifetime value (Resource /1/0/1) equal to 60 and send AT#LWM2MREG=0,2,99 on the host side to take effect
The host application receives many REGISTERING/REGISTERED messages from client	1. The registration update is not received by the Server due to slow network/packet loss	1. No action is needed, the server establishes by himself a new connection.

Problem	Cause	Action
After the XML file is injected on the modem file system, the object is not present on the object browser of OneEdge IoT Portal	1. The modem has not been rebooted 2. The file on the file system is corrupted	1. Send AT#REBOOT and restart LwM2M client 2. Enable Flow control on both host application/TATC and modem side (AT&K3)

Table 3: self diagnostic

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- it can interfere with other electronic devices, particularly in environments such as hospitals, airports, aircrafts, etc.
- there is a risk of explosion such as gasoline stations, oil refineries, etc. It is the responsibility of the user to enforce the country regulation and the specific environment regulation.

Do not disassemble the product; any mark of tampering will compromise the warranty validity. We recommend following the instructions of the hardware user guides for correct wiring of the product. The product has to be supplied with a stabilized voltage source and the wiring has to be conformed to the security and fire prevention regulations. The product has to be handled with care, avoiding any contact with the pins because electrostatic discharges may damage the product itself. Same cautions have to be taken for the SIM, checking carefully the instruction for its use. Do not insert or remove the SIM when the product is in power saving mode.

The system integrator is responsible for the functioning of the final product. Therefore, the external components of the module, as well as any project or installation issue, have to be handled with care. Any interference may cause the risk of disturbing the GSM network or external devices or having an impact on the security system. Should there be any doubt, please refer to the technical documentation and the regulations in force. Every module has to be equipped with a proper antenna with specific characteristics. The antenna has to be installed carefully in order to avoid any interference with other electronic devices and has to guarantee a minimum distance from the body (20 cm). In case this requirement cannot be satisfied, the system integrator has to assess the final product against the SAR regulation.

The equipment is intended to be installed in a restricted area location.

The equipment must be supplied by an external specific limited power source in compliance with the standard EN 62368-1.

The European Community provides some Directives for the electronic equipment introduced on the market. All of the relevant information is available on the European Community website:

https://ec.europa.eu/growth/sectors/electrical-engineering_en

7. GLOSSARY

ADC	Analog – Digital Converter
CA	Connection Agent
CLK	Clock
CMOS	Complementary Metal – Oxide Semiconductor
CS	Chip Select
DAC	Digital – Analog Converter
DTE	Data Terminal Equipment
ESR	Equivalent Series Resistance
EVB	Evaluation Board
EVK	Evaluation Kit
GPIO	General Purpose Input Output
HS	High Speed
HSDPA	High Speed Downlink Packet Access
HSIC	High Speed Inter Chip
HSUPA	High Speed Uplink Packet Access
I2C	Inter Integrated Circuit
I/O	Input Output
MCU	MicroController Unit
MISO	Master Input – Slave Output
MOSI	Master Output – Slave Input
MRDY	Master Ready
OMA	Open Mobile Alliance
PCB	Printed Circuit Board
PDP	Packet Data Protocol
RSSI	Radio Signal Strength Information
RTC	Real Time Clock
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready

TATC	Telit AT Controller
TTSC	Telit Technical Support Centre
UART	Universal Asynchronous Receiver Transmitter
UMTS	Universal Mobile Telecommunication System
URI	Universal Resource Identifier
URC	Unsolicited Result Code
USB	Universal Serial Bus

8. RELATED DOCUMENTS

[01]	1VV0301351	ME910C1 HW User Guide
[02]	1VV0301593	ME910G1 HW Design Guide
[03]	1VV0301588	ME310G1 HW Design Guide
[04]	80529ST10815A	ME910C1 AT Command Reference Guide
[05]	80617ST10991A	MEx10G1 AT Command Reference Guide
[06]	80502ST10950A	LE910Cx AT Command Reference Guide
[07]	80529ST10974A	ME910C1 LwM2M AT commands Reference Guide:
[08]	80617ST11022A	MEx10G1 LwM2M AT commands Reference Guide
[09]	80646ST11059A	LE910Cx LwM2M AT commands Reference Guide
[10]	80000NT11696A	2G 3G 4G Registration Process
[11]	1VV0301586	Connection Agent User Guide
[12]	V1_0_2-20180209-A	OMA Lightweight M2M Specification

9. DOCUMENT HISTORY

Revision	Date	Changes
12	2022-06-14	Editorial changes
11	2022-05-16	Updated some links related to new Telit web site; Related documents section moved in the end of the document according to new Telit template
10	2022-03-11	Removed SPI reference from hosted architecture in paragraph 2.2.1; added Connection Agent example in paragraph 4.2; modified Figure 12 in paragraph 4.2
9	2022-03-02	Some cosmetic improvements related to figure numbering; added support for LE910Cx devices; improved data connection section
8	2021-05-19	Self-Diagnostic Chapter added, updated Diagrams, create/delete custom object section added
7	2020-12-16	New Template applied
6	2020-07-01	Removed ACK mode; section "Note" moved to section "OneEdge preconditions"; converted code images to text snippets
5	2019-09-06	Set Wakeup SMS section added
4	2019-08-08	Updated "Registration Update" transmission policy, Note section added
3	2019-07-17	CA logic flow added, ACK mode added, Hostless cases (AppZone code) added in chapter 5
2	2019-04-16	Updated AppZone Diagrams
1	2019-04-04	Updated commands syntax
0	2019-01-31	First issue

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